

PROBLEM #004

Mathematics

Real Numbers & Inequalities

★★★ ■ INTERMEDIATE

SUBJECT

Mathematics — Algebra

LEVEL

Grade 10-12 / Olympiad

TOPICS

Real Numbers · Completing the Square · Inequalities

// PAF SCORING

PAF scores will be given based on the solution. Partial credit awarded for significant progress.

THE PROBLEM

Let x be a real number. Define the function:

$$f(x) = x^2 - x + 1$$

PART

a**PROVE $f(x) > 0$ FOR ALL REAL x** Show that $f(x) = x^2 - x + 1 > 0$ for every real number x .

Your proof must be complete and rigorous.

(a)

PART

b**NO REAL SOLUTIONS**

Using part (a), prove that the equation below has NO real solutions:

$$x^2 - x^2 + 1 = 0$$

Hint: consider the substitution $u = x^2 - x$.

(b)

PART

c**PROVE AN INEQUALITY**Let a, b be real numbers such that $a + b = 1$.

Using part (a), prove that:

$$a^2 + ab + b^2 \geq 3/4$$

State clearly where you use the result of part (a).

(c)

PART

d**FIND ALL REAL SOLUTIONS**Find all real numbers x satisfying:

$$x^2 - x + 1 = 1 / (x^2 - x + 2)$$

Justify that your solution(s) are real.

Hint: let $u = x^2 - x + 1$ and use part (a).

(d)